Beta Mixtures Across Integer Twists

Finite spectral corrections, harmonic logarithmic moments,
and Stieltjes–polylogarithm primitives

Research continuation prepared for the ProveIt project

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Abstract

The ProveIt manuscript evaluates convolutions of unequal twisted Hurwitz kernels by Dirichlet beta mixtures when the twists lie in one interval between consecutive integers. We determine the correction when that hypothesis is removed and the total order is a positive integer. For any finite number of factors, the difference between the convolution and the affine finite-part mixture is an explicit finite Fourier polynomial. Its coefficient at a crossed mode is a sine of the sum of the orders on the negative side. Although this correction vanishes at every positive-integer order tuple, its balanced order derivatives need not vanish. We give all such derivatives and prove that their first-order rank is one less than the number of occupied integer chambers.

A two-endpoint specialization evaluates every sufficiently vanishing centered-logarithm moment by a polynomial times $\pi u \cot \pi u$. At integer endpoint orders the resulting ordinary integrals are finite rational linear combinations of even zeta values. Their parameter primitives have finite classical-polylogarithm expressions, producing convergent inverse-logarithm integrals with odd zeta values. An independent logistic Mellin transform gives finite Hurwitz-zeta reductions, all logarithmic finite parts, and generalized Stieltjes antiderivatives. In particular, the finite part of an unnormalized Mellin integral requires an explicit Gamma-derivative contribution from the pole residue.

The proofs are self-contained apart from standard special-function identities, which are also derived where used. The package records 508 finite exact checks, including three negative controls, and 94 floating-point diagnostics. These checks are not a formalization or interval certification. The contribution is an extension of the inspected repository material, not a claim of literature priority.

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1 Research target, relationship to the manuscript, and scope

The immediate starting point is the manuscript section *Unequal twists through beta mixtures and finite resolvent identities*, file `07-mixed-twist-identities.tex` [1]. Its kernel K_α^θ has Fourier multiplier $(2\pi i(n+\theta))^{-\alpha}$. When all real twists belong to a single open interval between consecutive integers, its Dirichlet-mixture theorem is valid for arbitrary orders with positive real parts. The same section gives finite partial-fraction reductions at positive integer orders without the common-interval restriction. Both statements are correct as written.

There is a genuine gap between those regimes: the beta simplex can cross an integral twist even when none of its vertices is integral. The intermediate kernel then has a pole in the twist parameter. Simply replacing the integral by a principal value, or by the real finite part at higher order, does not in general preserve the identity. Moreover, an identity valid only on the discrete set of integer order tuples cannot be differentiated in the orders without first constructing its analytic extension.

The main theorem below fills this gap at every positive integer *total* order, allowing the individual orders to be fractional. It is not restricted to two factors. The resulting correction is especially simple because only finitely many Fourier modes can change sign. This finite-dimensional feature makes all balanced order derivatives explicit and permits a complete rank calculation. The same scalar computation produces a family of exact ordinary integrals, rather than estimates.

1.1 What is established here

The central results are Theorem 4.1 (arbitrary-depth integer-total crossing formula), Theorem 5.1 (all balanced order derivatives), and Theorem 5.3 (the complete chamberwise criterion for their vanishing). The analytic consequences are Theorem 6.1, the polylogarithm primitive in Proposition 7.1, the logarithmic Mellin finite-part formula in Theorem 9.1, and the Stieltjes primitive ladder in Theorem 9.3. Every one is proved below.

The beta integral, Gamma reflection, integer polygamma–Hurwitz conversion, elementary polylogarithm differential rule, and finite Hurwitz expansion of the equal-period Barnes-type zeta are classical ingredients [4, 5, 6, 7, 8]. Their use is not itself a novelty claim. The distinction between analytic boundary values and uncorrected Feynman-parameter integrals also has a substantial history; the papers of Weldon and Bedaque–Das provide relevant context [9, 10]. Only their abstracts were consulted for this contextual comparison. No step of our proof is outsourced to those papers, and no assertion is made that the general boundary-value principle originates here.

1.2 What is not claimed

This is not a proof of the manuscript's S_6 or revised S_8 period conjectures. The repository README continues to mark them conjectural [2]. Nor do we claim a finite reduction of all arbitrary unequal-twist order derivatives to a prescribed list of single Hurwitz values. Our rank theorem concerns explicit Fourier corrections, not arithmetic independence of periods.

The targeted source was read again at the reference commit printed on the title page. The manuscript README, the chapter inventory, and the incoming directory listing were inspected. The harmonic Gamma-coefficient section was read in full; it already proves the corrected all-order Choi families [3]. Only the beginning of the log-Gamma-moments section was inspected. The six binary archives present in the incoming listing were *not* opened. Consequently this is neither an exhaustive repository audit nor an exhaustive comparison with all pending reports. The machine-readable provenance record makes these boundaries explicit.

2 Fourier, branch, and finite-part conventions

Write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. For a periodic distribution f, use

$$\widehat{f}(n) = \langle f, \mathrm{e}^{-2\pi i n x} \rangle, \qquad f = \sum_{n \in \mathbb{Z}} \widehat{f}(n) \mathrm{e}^{2\pi i n x}.$$

The pairing agrees with integration over $[0,1]$ for integrable functions. Convolution is normalized so that Fourier coefficients multiply. For a real nonintegral θ, define the entire distribution-valued family

$$\widehat{K_\alpha^\theta}(n) = \exp\left[-\alpha\left\{\log(2\pi|n+\theta|) + \frac{\pi i}{2}\operatorname{sgn}(n+\theta)\right\}\right]. \tag{2.1}$$

On each compact set of α, these coefficients and their order derivatives have a common polynomial growth bound in n. Pairing against a smooth test function therefore proves normal convergence and entire dependence on α. The same argument defines convolution of any finite collection of these kernels.

For positive integers m and $0 < x < 1$,

$$K_1^\theta(x) = \frac{\mathrm{e}^{-2\pi i\theta x}}{1-\mathrm{e}^{-2\pi i\theta}}, \qquad K_m^\theta(x) = \frac{(-1)^{m-1}}{(2\pi i)^{m-1}(m-1)!}\partial_\theta^{m-1} K_1^\theta(x). \tag{2.2}$$

Indeed the first expression has Fourier coefficient $(2\pi i(n+\theta))^{-1}$, by a direct integral. Differentiating this coefficient gives the second expression. Endpoint representatives are not assigned by this formula; the periodic distribution is the primary object.

2.1 The affine finite part

For a positive integer m we use the distributional identity

$$(y - i0)^{-m} = \operatorname{Pf} y^{-m} + \frac{i\pi(-1)^{m-1}}{(m-1)!}\delta^{(m-1)}(y). \tag{2.3}$$

Here $\operatorname{Pf} y^{-1}$ is the Cauchy principal value, and

$$\operatorname{Pf} y^{-m} = \frac{(-1)^{m-1}}{(m-1)!}\frac{d^{m-1}}{dy^{m-1}}\operatorname{PV}\frac{1}{y}.$$

For completeness, write $(y-i\varepsilon)^{-1}$ as $y/(y^2+\varepsilon^2) + i\varepsilon/(y^2+\varepsilon^2)$. The first term tends to the principal value and the second to $i\pi\delta$. Differentiating $m-1$ times proves (2.3). Replacing $i0$ by $-i0$ reverses the sign of the delta term.

If h is smooth near zero and integrable away from zero on a compact interval, write

$$\operatorname{FP}\int h(y) y^{-m}\, dy = \langle \operatorname{Pf} y^{-m}, h\rangle.$$

We also allow a finite measure away from zero; only its restriction near zero needs a smooth density. A partition of unity reduces this definition to a smooth function of compact support near zero and an ordinary integral against the measure elsewhere. An explicit calculation uses Taylor subtraction through degree $m-1$ near zero, followed by the exact finite parts of the subtracted monomials.

Thus the notation specifies a computable normalization, not merely a rule to "discard infinity." Pairing (2.3) with h gives

$$\int h(y)(y-i0)^{-m}\,dy = \mathrm{FP}\int h(y)y^{-m}\,dy + \frac{i\pi}{(m-1)!}h^{(m-1)}(0). \tag{2.4}$$

All simplex finite parts below use the displayed *affine scalar* $y=\sum t_j y_j$. This choice matters: an arbitrary nonlinear change of variable can alter local finite-part terms. Later, logarithmic Mellin finite parts use a different, explicitly stated cutoff variable. We change variables between the two settings only for ordinary convergent integrals, unless a separate normalization is supplied.

3 A scalar Dirichlet crossing identity at every depth

Let

$$\Delta_{d-1}=\{(t_1,\dots,t_d): t_j\ge 0,\ \sum_j t_j=1\},\qquad dt=dt_1\cdots dt_{d-1},$$

where $t_d=1-\sum_{j<d}t_j$. Fix nonzero real numbers y_j and complex parameters α_j satisfying

$$\mathrm{Re}\,\alpha_j>0,\qquad \sum_{j=1}^{d}\alpha_j=m\in\mathbb{N}. \tag{3.1}$$

Set

$$\sigma=\sum_{y_j<0}\alpha_j,\qquad Q=\prod_{j=1}^{d}|y_j|^{-\alpha_j}. \tag{3.2}$$

The normalized pushforward h is specified by

$$\int h(y)F(y)\,dy=\frac{\Gamma(m)}{\prod_j\Gamma(\alpha_j)}\int_{\Delta_{d-1}}\prod_j t_j^{\alpha_j-1}F\left(\sum_j t_jy_j\right)dt. \tag{3.3}$$

For complex parameters this is a complex measure, not a probability density. Positivity is unnecessary.

Lemma 3.1 (Smoothness at an interior crossing)**.** *If the y_j include both signs, the pushforward in (3.3) has a smooth density on a neighborhood of zero. Every fixed derivative there depends holomorphically on the parameters in (3.1). If all signs agree, its support is bounded away from zero.*

Proof. Group the negative and positive indices. The usual change of simplex coordinates first chooses the total negative mass v, then the internal coordinates in each group. It factors the normalized measure into two internal Dirichlet measures and the beta density

$$\frac{\Gamma(m)}{\Gamma(\sigma)\Gamma(m-\sigma)}v^{\sigma-1}(1-v)^{m-\sigma-1}\,dv.$$

This factorization follows directly by its Jacobian and the beta integral; it remains valid for positive real parts by absolute convergence. Conditional on the internal coordinates, let Y_- and Y_+ be the corresponding weighted means of the negative and positive y_j. Some $c>0$ satisfies $Y_-\le -c$ and $Y_+\ge c$ uniformly. Now $y=vY_-+(1-v)Y_+$, so

$$v=\frac{Y_+-y}{Y_+-Y_-},\qquad \left|\frac{dv}{dy}\right|=\frac{1}{Y_+-Y_-}.$$

For $|y| < c/2$, both v and $1 - v$ stay uniformly away from zero, because $Y_\pm$ also range in fixed compact sets. Every y derivative of the conditional beta density is therefore bounded by a constant times an integrable internal Dirichlet weight. Differentiation under the integral proves smoothness. The same domination on compact parameter sets, with additional endpoint logarithms when needed, proves holomorphic dependence. The one-sign assertion is immediate from convexity. □

Theorem 3.2 (Finite part and crossing derivative)**.** *Under* (3.1)*,*

$$\mathrm{FP} \int h(y) y^{-m}\, dy = \cos(\pi\sigma) Q, \tag{3.4}$$

$$h^{(m-1)}(0) = \frac{(m-1)!}{\pi} \sin(\pi\sigma) Q. \tag{3.5}$$

In the one-sign case the second formula is read as zero, since $\sigma = 0$ or $\sigma = m$. The two boundary values are

$$\int h(y)(y \mp i0)^{-m}\, dy = \mathrm{e}^{\pm i\pi\sigma} Q. \tag{3.6}$$

Proof. For complex numbers z_j in the right half-plane, the product of Gamma integrals and the substitution $u_j = r t_j$ give

$$\prod_j z_j^{-\alpha_j} = \frac{\Gamma(m)}{\prod_j \Gamma(\alpha_j)} \int_{\Delta_{d-1}} \frac{\prod_j t_j^{\alpha_j - 1}}{(\sum_j t_j z_j)^m}\, dt. \tag{3.7}$$

Indeed the Jacobian is r^{d-1} and integration in r produces $\Gamma(m)(\sum t_j z_j)^{-m}$. This proves the identity directly in a domain where all relevant integrals converge absolutely.

Put $z_j = \varepsilon + i y_j$ in (3.7) and factor out i^{-m} from both sides. As $\varepsilon \downarrow 0$, the left side becomes $Q\mathrm{e}^{i\pi\sigma}$. For a negative y_j the factor $(y_j - i0)^{-\alpha_j}$ contributes $\mathrm{e}^{i\pi\alpha_j}$; this is the entire phase correction. On the right, Lemma 3.1 and (2.4) justify the distributional limit at zero; away from zero dominated convergence applies to the finite pushforward measure. This proves the upper sign of (3.6). The substitution $z_j = \varepsilon - i y_j$ proves the lower sign by the same argument. Average the two identities to obtain (3.4); subtract them and use (2.4) to obtain (3.5). No reality assumption on the parameters is needed in this subtraction. □

Remark 3.3 (A direct two-point version). For $A, B > 0$, $\mathrm{Re}\,\alpha, \mathrm{Re}\,\beta > 0$ and $\alpha + \beta = m$, the theorem states

$$\frac{\Gamma(m)}{\Gamma(\alpha)\Gamma(\beta)} \mathrm{FP} \int_0^1 \frac{t^{\alpha-1}(1-t)^{\beta-1}}{[B - (A+B)t]^m}\, dt = \frac{\cos(\pi\alpha)}{A^\alpha B^\beta}. \tag{3.8}$$

The missing imaginary boundary term is $i\sin(\pi\alpha)/(A^\alpha B^\beta)$. Here the affine singularity is $t = B/(A+B)$, not necessarily $1/2$.

4 The finite spectral correction for unequal twists

For nonintegral real twists $\theta_1, \ldots, \theta_d$, define

$$\mathcal{C} = \{n \in \mathbb{Z} : \min_j (n + \theta_j) < 0 < \max_j (n + \theta_j)\}. \tag{4.1}$$

This is a finite set. For every mode put

$$\sigma_n = \sum_{n+\theta_j<0} \alpha_j, \qquad Q_n = \prod_j |n+\theta_j|^{-\alpha_j}.$$

Let $\mathcal{M}_m(\boldsymbol{\alpha};\boldsymbol{\theta})$ denote the simplex mixture whose nth Fourier coefficient is the affine finite part of

$$\frac{1}{(2\pi i)^m}\frac{\Gamma(m)}{\prod_j \Gamma(\alpha_j)} \int_{\Delta_{d-1}} \prod_j t_j^{\alpha_j-1} \left(n+\sum_j t_j\theta_j\right)^{-m} dt.$$

Equivalently it is the affine finite-part interpretation of the integral of $K_m^{\sum t_j\theta_j}$. Only finitely many modes need subtraction.

Theorem 4.1 (Arbitrary-depth integer-total crossing formula)**.** *If* $\operatorname{Re}\alpha_j > 0$ *and* $\sum_j \alpha_j = m \in \mathbb{N}$*, then*

$$K_{\alpha_1}^{\theta_1} * \cdots * K_{\alpha_d}^{\theta_d} = \mathcal{M}_m(\boldsymbol{\alpha};\boldsymbol{\theta}) + \frac{i}{(2\pi i)^m} \sum_{n\in\mathcal{C}} \mathrm{e}^{2\pi i n x} \sin(\pi\sigma_n) Q_n. \tag{4.2}$$

The mixture and its correction are holomorphic in the indicated parameter hyperplane. If all twists lie in one open unit interval, the correction is empty. If every α_j *is a positive integer, the correction vanishes even when* $\mathcal{C}$ *is nonempty.*

Proof. At mode n, formula (2.1) gives the product multiplier

$$\prod_j (2\pi i(n+\theta_j))^{-\alpha_j} = (2\pi i)^{-m} \mathrm{e}^{i\pi\sigma_n} Q_n. \tag{4.3}$$

Theorem 3.2 evaluates the mixture coefficient as $(2\pi i)^{-m}\cos(\pi\sigma_n)Q_n$. Their difference is the displayed sine term. For $n \notin \mathcal{C}$, σ_n is either 0 or m, so the difference is zero. This also proves the assertions about the common interval and integer tuples.

For $|n|$ sufficiently large all $n+\theta_j$ have the same sign and are comparable to $|n|$. Uniformly on compact parameter sets in the hyperplane, $Q_n = O(|n|^{-m})$. The finitely many crossed coefficients are holomorphic by Lemma 3.1. Pairing the series with a smooth test function now justifies the mixture, all parameter derivatives, and the equality of distributions by Fourier uniqueness. In fact at $m \geq 2$ the undifferentiated coefficients are absolutely summable. The distributional formulation also covers $m = 1$ without any endpoint convention. □

The correction in (4.2) is a finite trigonometric polynomial in x. The local delta derivative used to obtain it was in the *intermediate twist variable*. It must not be replaced by a Dirac distribution in x.

Corollary 4.2 (Two twists crossing several integers)**.** *For nonintegral* $a < b$ *and* $\alpha+\beta = m$ *with positive real parts,*

$$\begin{aligned} K_\alpha^a * K_\beta^b = {} & \frac{\Gamma(m)}{\Gamma(\alpha)\Gamma(\beta)} \operatorname{FP}\int_0^1 (1-t)^{\alpha-1} t^{\beta-1} K_m^{(1-t)a+tb}\, dt \\ & + \frac{i\sin(\pi\alpha)}{(2\pi i)^m} \sum_{\substack{k\in\mathbb{Z}\\ a<k<b}} \frac{\mathrm{e}^{-2\pi i k x}}{(k-a)^\alpha (b-k)^\beta}. \end{aligned} \tag{4.4}$$

Proof. The crossed modes are exactly $n = -k$ with $a < k < b$. Their negative group consists of the first factor, so $\sigma_{-k} = \alpha$. □

4.1 Exact counterexamples to an uncorrected extension

At $a = -1/2$, $b = 1/2$ and $\alpha = \beta = 1/2$, only $n = 0$ crosses. Hence

$$K_{1/2}^{-1/2} * K_{1/2}^{1/2} = \frac{1}{\pi}\,\mathrm{PV}\int_0^1 \frac{K_1^{t-1/2}}{\sqrt{t(1-t)}}\,dt + \frac{1}{\pi}. \tag{4.5}$$

The zero Fourier coefficient of the convolution is $1/\pi$, whereas that of the principal-value mixture is zero. Thus even a single mode disproves the unrestricted, uncorrected beta formula. This is an exact counterexample, independent of floating-point calculations.

For a genuinely three-factor example, let the twists be $(-1/2, 1/2, 3/2)$ and all three orders be $1/3$. The total order is one; the crossed modes are 0 and -1. Both have $Q_n = 2\,3^{-1/3}$ and $\sin(\pi\sigma_n) = \sqrt{3}/2$. Therefore the correction is

$$\frac{3^{1/6}}{2\pi}\left(1 + \mathrm{e}^{-2\pi i x}\right). \tag{4.6}$$

Repeated twists cause no difficulty: identical vertices may be grouped by adding their orders, directly from the beta integral or the Fourier multipliers.

5 All balanced order jets and their chamber rank

Fix positive integers r_j with $\sum_j r_j = m$. Choose a direction $\boldsymbol{v} = (v_1, \ldots, v_d)$ with $\sum_j v_j = 0$, and put $\alpha_j = r_j + u v_j$. For sufficiently small u the hypotheses of Theorem 4.1 hold. At a crossed mode define

$$\begin{aligned} R_n &= \sum_{n+\theta_j<0} r_j, & V_n &= \sum_{n+\theta_j<0} v_j, \\ L_n &= -\sum_j v_j \log|n+\theta_j|, & Q_{n,0} &= \prod_j |n+\theta_j|^{-r_j}. \end{aligned} \tag{5.1}$$

For a nonnegative integer q, call C_q the qth u derivative at zero of the correction in (4.2).

Theorem 5.1 (Every balanced derivative, without a recursion)**.** *For every $q \geq 0$,*

$$C_q(x) = \frac{1}{2(2\pi i)^m} \sum_{n\in\mathcal{C}} (-1)^{R_n} Q_{n,0} \mathrm{e}^{2\pi i n x} \left\{(L_n + i\pi V_n)^q - (L_n - i\pi V_n)^q\right\}. \tag{5.2}$$

In particular $C_0 = 0$, but

$$C_1(x) = \frac{i\pi}{(2\pi i)^m} \sum_{n\in\mathcal{C}} (-1)^{R_n} Q_{n,0} V_n \mathrm{e}^{2\pi i n x}. \tag{5.3}$$

Proof. At each crossed mode the varying factor is

$$i\sin(\pi(R_n + uV_n))\, Q_{n,0}\mathrm{e}^{uL_n} = \frac{(-1)^{R_n} Q_{n,0}}{2}\left(\mathrm{e}^{u(L_n + i\pi V_n)} - \mathrm{e}^{u(L_n - i\pi V_n)}\right).$$

Differentiate this finite sum. Holomorphic dependence in the preceding theorem justifies differentiating the convolution and mixture as well. □

For several balanced directions $\boldsymbol{v}^{(1)},\dots,\boldsymbol{v}^{(p)}$, replace α_j by $r_j+\sum_\ell u_\ell v_j^{(\ell)}$. Define $L_{n,\ell}$ and $V_{n,\ell}$ as in (5.1). The mixed derivative of multi-order $\boldsymbol{q}$ has the same formula with the brace replaced by

$$\prod_{\ell=1}^{p}(L_{n,\ell}+i\pi V_{n,\ell})^{q_\ell}-\prod_{\ell=1}^{p}(L_{n,\ell}-i\pi V_{n,\ell})^{q_\ell}. \tag{5.4}$$

Thus arbitrarily high mixed derivatives require only finite polynomials in logarithms and π on the correction side.

5.1 Harmonic-number weights on the simplex

Let $\mathcal{Y}_q$ denote the complete exponential Bell polynomial, normalized by

$$\exp\left(\sum_{k\ge1}\ell_k\frac{u^k}{k!}\right)=\sum_{q\ge0}\mathcal{Y}_q(\ell_1,\dots,\ell_q)\frac{u^q}{q!}.$$

For $N\ge0$ and $k\ge1$, put $H_N^{(k)}=\sum_{j=1}^{N}j^{-k}$, with $H_0^{(k)}=0$ and $H_N=H_N^{(1)}$. The simplex density at the integer tuple is

$$B_0(\boldsymbol{t})=\frac{\Gamma(m)}{\prod_j\Gamma(r_j)}\prod_j t_j^{r_j-1}.$$

Its qth balanced derivative is

$$B_0(\boldsymbol{t})\mathcal{Y}_q(\ell_1,\dots,\ell_q), \tag{5.5}$$

where

$$\ell_1=\sum_j v_j(\log t_j-H_{r_j-1}), \tag{5.6}$$

$$\ell_k=(-1)^{k+1}(k-1)!\sum_j v_j^k(\zeta(k)-H_{r_j-1}^{(k)}),\qquad k\ge2. \tag{5.7}$$

To prove these expressions, differentiate the logarithm of the normalized Dirichlet density. Its first derivative is $\sum_j v_j(\log t_j-\psi(r_j))$. The Euler constant cancels since $\sum v_j=0$ and $\psi(r_j)=H_{r_j-1}-\gamma$. Higher derivatives are $-\sum_j v_j^k\psi^{(k-1)}(r_j)$. The convergent polygamma series gives (5.7). Exponentiation proves (5.5).

Consequently the derivative of the mixture is the affine finite-part integral of $K_m^{\sum t_j\theta_j}$ against (5.5). These formulas evaluate whole families of logarithmic simplex integrals. They are not merely derivatives of an identity at an isolated integer point: the required analytic family has already been established.

Example 5.2 (A zero correction with a nonzero derivative)**.** For $\theta_1=-1/2$, $\theta_2=1/2$ and $(\alpha_1,\alpha_2)=(1+u,1-u)$, the zero-mode coefficient of the convolution is $\pi^{-2}\mathrm{e}^{i\pi u}$. The corresponding finite-part coefficient is $\pi^{-2}\cos(\pi u)$. Their difference is

$$\frac{i\sin(\pi u)}{\pi^2}. \tag{5.8}$$

It is zero at $u=0$, but its first and third derivatives there are i/π and $-i\pi$. The second derivative is zero. An integer-order partial-fraction formula alone cannot supply these order derivatives.

5.2 Exactly which directions have no correction?

Let c be the number of occupied open unit intervals $(k, k+1)$ among the twists, and let $I_1, \dots, I_c$ be their groups, ordered from left to right. Empty intervals are not counted.

Theorem 5.3 (Chamberwise kernel and rank)**.** *On the balanced tangent space $\{\boldsymbol{v} : \sum_j v_j = 0\}$, the linear map $\boldsymbol{v} \mapsto C_1$ has rank $c-1$. Its kernel is exactly*

$$\left\{ \boldsymbol{v} : \sum_{j \in I_h} v_j = 0 \text{ for every } h = 1, \dots, c \right\}, \tag{5.9}$$

and has dimension $d-c$. A direction belongs to this kernel if and only if every *C_q vanishes. In that case the full correction vanishes identically as a function of u near zero.*

Proof. Because the exponential functions at distinct integer frequencies are linearly independent and $Q_{n,0} \neq 0$, formula (5.3) vanishes if and only if $V_n = 0$ at each crossed mode. For $n = -k$, the negative group is $\{j : \theta_j < k\}$. It is a union of complete occupied chambers. As the integer k runs between the smallest and largest twists, these groups realize every nonempty proper prefix $I_1 \cup \cdots \cup I_h$, $1 \le h < c$. Gaps between occupied chambers repeat a prefix but do not create an additional one.

Writing $s_h = \sum_{j \in I_h} v_j$, the distinct conditions are $s_1 = 0$, $s_1 + s_2 = 0$, …, $s_1 + \cdots + s_{c-1} = 0$. Together with $\sum_h s_h = 0$, these are equivalent to all $s_h = 0$. The map from $\boldsymbol{v}$ to the chamber sums is onto the $(c-1)$-dimensional zero-sum space: choose one representative index per chamber. Hence the rank is $c-1$ and the kernel dimension is $d-c$. If all V_n are zero, the sine factor $\sin(\pi(R_n + uV_n))$ is identically zero; thus every derivative vanishes. The converse already follows from $q = 1$. □

For example, twists $(1/5, 7/10, 6/5, 18/5)$ occupy three chambers. The rank is two and the kernel is generated by $(1, -1, 0, 0)$. This rank is purely spectral and exact. It gives no information about linear independence of special values obtained by evaluating the kernels at a point.

6 Centered logarithmic moments and finite even-zeta reductions

The same crossing calculation produces ordinary real integrals with arbitrary logarithmic order. Fix $A, B > 0$ and define

$$U(t) = \log \frac{At}{B(1-t)}, \qquad D(t) = B - (A+B)t, \qquad t_0 = \frac{B}{A+B}. \tag{6.1}$$

Both U and D vanish simply at t_0. We initially allow positive real r, s such that $r + s = m \in \mathbb{N}$. The stronger polynomial conclusions will assume that r, s themselves are positive integers.

Theorem 6.1 (The centered generator and all ordinary moments)**.** *For $-r < \operatorname{Re} u < s$,*

$$I_{r,s}(u) := \operatorname{FP} \int_0^1 \frac{t^{r-1}(1-t)^{s-1} \mathrm{e}^{uU(t)}}{D(t)^m}\, dt = \frac{\Gamma(r+u)\Gamma(s-u)\cos(\pi(r+u))}{(m-1)! A^r B^s}. \tag{6.2}$$

The finite part is in the affine variable D. If $r, s \in \mathbb{N}$, put

$$P_{r,s}(u) = \prod_{j=1}^{r-1}(j+u) \prod_{j=1}^{s-1}(j-u), \qquad G_{r,s}(u) = \pi u \cot(\pi u) P_{r,s}(u). \tag{6.3}$$

Empty products equal one. Then

$$I_{r,s}(u) = \frac{(-1)^r}{(m-1)!A^rB^s} G_{r,s}(u). \tag{6.4}$$

For every integer $q \geq m$, the following integral is ordinary and absolutely convergent:

$$\begin{aligned} M_q^{r,s}(A,B) &:= \int_0^1 \frac{t^{r-1}(1-t)^{s-1}U(t)^q}{D(t)^m}\,dt \\ &= \frac{2(-1)^{r+1}q!}{(m-1)!A^rB^s} \sum_{k=1}^{\lfloor q/2 \rfloor} \zeta(2k)[u^{q-2k}]P_{r,s}(u). \end{aligned} \tag{6.5}$$

Proof. Since $\mathrm{e}^{uU(t)} = (A/B)^u t^u (1-t)^{-u}$, apply (3.8) with $\alpha = r+u$ and $\beta = s-u$. The factors $A^{-u}B^u$ on its right cancel the centering factor $(A/B)^u$. This proves (6.2). Endpoint integrability on compact subsets of the indicated strip, and Taylor subtraction at t_0, show that both sides are holomorphic there.

For integer r, s, the Gamma recurrence and reflection identity give

$$\Gamma(r+u)\Gamma(s-u) = P_{r,s}(u)\Gamma(1+u)\Gamma(1-u) = P_{r,s}(u)\frac{\pi u}{\sin \pi u}.$$

Because $\cos(\pi(r+u)) = (-1)^r \cos \pi u$, this proves (6.4). The apparent poles at the nonzero integers inside the strip are canceled by $P_{r,s}$; the value at zero is removable.

Near t_0, $U(t)^q/D(t)^m = O(|t-t_0|^{q-m})$. For $q \geq m$ this is bounded, and the endpoint logarithms are integrable because $r, s > 0$. Thus the finite-part derivative $I_{r,s}^{(q)}(0)$ equals the ordinary integral in (6.5). Finally, the logarithmic derivative of the product for $\sin \pi u$, or the partial fractions of cotangent, gives the convergent expansion

$$\pi u \cot \pi u = 1 - 2\sum_{k \geq 1} \zeta(2k) u^{2k}, \qquad |u| < 1. \tag{6.6}$$

For example, the partial fractions are $\pi u \cot \pi u = 1 - 2\sum_{n \geq 1} u^2/(n^2 - u^2)$, and expanding the absolutely convergent right side proves (6.6). Since $\deg P_{r,s} = m - 2 < q$, the initial polynomial contributes no u^q coefficient. Coefficient extraction proves the stated formula. □

The endpoint asymmetry has disappeared from the answer except for the factor $A^{-r}B^{-s}$. The centered logarithm is essential for this simplification. An uncentered logarithm gives derivatives of an additional exponential in u.

6.1 Harmonic coefficients and parity

Every coefficient of $P_{r,s}$ is explicit in generalized harmonic numbers. Namely,

$$[u^j]P_{r,s}(u) = \frac{(r-1)!(s-1)!}{j!} \mathcal{Y}_j(p_1, \ldots, p_j), \tag{6.7}$$

where

$$p_k = (k-1)!\left((-1)^{k-1}H_{r-1}^{(k)} - H_{s-1}^{(k)}\right). \tag{6.8}$$

To verify this, take the logarithm of the two finite products in (6.3) and expand each $\log(1 \pm u/j)$. This is an identity of finite Taylor coefficients near zero. Although the Bell expressions are written for every j, the coefficients vanish for $j > m-2$.

For $r = s$, $P_{r,r}(u) = \prod_{j=1}^{r-1}(j^2 - u^2)$ is even. Therefore all odd centered moments vanish. The same fact follows directly from $t \mapsto 1 - t$ when $A = B$, and from the logistic coordinate used below for arbitrary positive A, B.

Some explicit ordinary evaluations, with $A = B = 1$ and $U = \log(t/(1-t))$, are

$$\int_0^1 \frac{U^{2k}}{(1-2t)^2}\,dt = 2(2k)!\zeta(2k), \qquad k \geq 1, \tag{6.9}$$

$$\int_0^1 \frac{(1-t)U^3}{(1-2t)^3}\,dt = -\pi^2, \tag{6.10}$$

$$\int_0^1 \frac{(1-t)U^4}{(1-2t)^3}\,dt = \frac{4\pi^4}{15}, \tag{6.11}$$

$$\int_0^1 \frac{t(1-t)U^4}{(1-2t)^4}\,dt = 8\big(\zeta(2) - \zeta(4)\big), \tag{6.12}$$

$$\int_0^1 \frac{t(1-t)U^6}{(1-2t)^4}\,dt = 240\big(\zeta(4) - \zeta(6)\big). \tag{6.13}$$

For instance $P_{1,2} = 1 - u$ and $P_{2,2} = 1 - u^2$, so each evaluation is an immediate finite coefficient extraction. None uses a numerical integer-relation search.

6.2 An ordinary subtracted generating integral

Let $T_N F(u) = \sum_{q=0}^{N} F^{(q)}(0)u^q/q!$ be the Taylor polynomial at zero. Write $c_{r,s} = (-1)^r/((m-1)!A^rB^s)$. The finite-part generator can be converted into an ordinary one:

Corollary 6.2 (Subtraction of the full local obstruction)**.** *For $r, s \in \mathbb{N}$ and $-r < \operatorname{Re} u < s$,*

$$\int_0^1 \frac{t^{r-1}(1-t)^{s-1}}{D(t)^m}\left(\mathrm{e}^{uU(t)} - \sum_{q=0}^{m-1}\frac{u^qU(t)^q}{q!}\right)dt = c_{r,s}\big(G_{r,s}(u) - T_{m-1}G_{r,s}(u)\big). \tag{6.14}$$

The integral is ordinary, locally uniformly convergent in the strip.

Proof. At t_0 the numerator is $O(U^m)$, uniformly on compact u sets. At the endpoints, the exponential term is bounded by powers with positive integrability exponents on every compact substrip, and the subtractions have only logarithmic growth. Taylor subtraction in (6.2) therefore agrees with the ordinary integral on the left. Equation (6.4) gives the right side. □

7 Classical-polylogarithm primitives and inverse-logarithm integrals

The generating function in (6.4) has elementary meromorphic factors, but its antiderivatives naturally introduce polylogarithms. The following finite formula avoids a recursive chain of unevaluated integrals.

Proposition 7.1 (A finite primitive for every polynomial cotangent)**.** *Let R be a polynomial of degree d. On any sufficiently small simply connected domain avoiding the integers, choose consistent local branches of $\mathrm{Li}_j(\mathrm{e}^{2\pi i u})$. Then*

$$\mathcal{A}_R(u) = -i\pi \int R(u)\,du - \sum_{j=0}^{d} \frac{(-1)^j R^{(j)}(u)}{(2\pi i)^j}\,\mathrm{Li}_{j+1}(\mathrm{e}^{2\pi i u}) \tag{7.1}$$

is an antiderivative of $\pi R(u)\cot \pi u$.

Proof. Put $z = \mathrm{e}^{2\pi i u}$ and $c = 2\pi i$. Since $\mathrm{Li}_0(z) = z/(1-z)$, the elementary cotangent formula gives

$$\cot \pi u = -i\big(1 + 2\,\mathrm{Li}_0(z)\big).$$

The differential identities $\partial_u \mathrm{Li}_{j+1}(z) = c\,\mathrm{Li}_j(z)$ follow from the power series in $|z| < 1$ and then by analytic continuation on the chosen local branches. Differentiating the finite sum in (7.1), adjacent terms cancel. The only surviving polylogarithmic term is $-cR(u)\,\mathrm{Li}_0(z)$, while differentiating the polynomial primitive contributes $-i\pi R(u)$. Their sum is $\pi R(u)\cot \pi u$. □

This is a local primitive, not a globally single-valued function on the punctured plane. In particular the logarithm in $\mathrm{Li}_1(z) = -\log(1-z)$ requires a branch. For the definite real integrals below, analytic continuation from the upper half u-plane gives an unambiguous boundary value on $0 < u < 1$.

Corollary 7.2 (Normalized real primitive)**.** *Suppose R has real coefficients and $R(0) = 0$. For $0 < v < 1$,*

$$\begin{aligned}\int_0^v \pi R(u)\cot \pi u\,du = &-\operatorname{Re}\sum_{j=0}^{d} \frac{(-1)^j R^{(j)}(v)}{(2\pi i)^j}\,\mathrm{Li}_{j+1}(\mathrm{e}^{2\pi i v}) \\ &+\operatorname{Re}\sum_{j=1}^{d} \frac{(-1)^j R^{(j)}(0)}{(2\pi i)^j}\zeta(j+1).\end{aligned} \tag{7.2}$$

If additionally $v = 1/2$, this becomes the finite odd-zeta expression

$$\begin{aligned}\int_0^{1/2} \pi R(u)\cot \pi u\,du = {}& R(1/2)\log 2 \\ &+ \sum_{\ell=1}^{\lfloor d/2 \rfloor} \frac{(-1)^\ell}{(2\pi)^{2\ell}} \Big((1-2^{-2\ell})R^{(2\ell)}(1/2) + R^{(2\ell)}(0)\Big)\,\zeta(2\ell+1).\end{aligned} \tag{7.3}$$

Proof. The real part of $-i\pi \int R$ is zero along the real segment. As $u \downarrow 0$, the $j = 0$ term tends to zero since $R(u) = O(u)$ and $\mathrm{Li}_1(\mathrm{e}^{2\pi i u}) = O(1 + |\log u|)$. For $j \geq 1$, the polylogarithm tends to $\zeta(j+1)$. Subtract the endpoint value in Proposition 7.1. This proves (7.2). At $v = 1/2$, use $\mathrm{Li}_1(-1) = -\log 2$ and $\mathrm{Li}_{j+1}(-1) = -(1-2^{-j})\zeta(j+1)$ for $j \geq 1$. The odd-j terms have zero real part, giving (7.3). □

For $R(u) = uP_{r,s}(u)$, the degree is $m-1$ and $\pi R \cot \pi u = G_{r,s}$. Thus polylogarithms through order m suffice for the primitive of every centered generator. In the symmetric example $r = s = 2$, $R(u) = u - u^3$, and

$$\int_0^{1/2} \pi(u-u^3)\cot \pi u\,du = \frac{3}{8}\log 2 + \frac{9\zeta(3)}{16\pi^2}. \tag{7.4}$$

The appearance of odd zeta values in a primitive is compatible with the even-zeta Taylor coefficients of the original generator: integration and evaluation at a nonzero endpoint are different operations.

7.1 An all-order ordinary inverse-logarithm family

Integrating (6.14) in the parameter, rather than in t, gives the following exact identity.

Theorem 7.3 (Convergent inverse-logarithm integrals)**.** *Let $r, s \in \mathbb{N}$, $m = r + s$, and $0 < v < 1$. Then*

$$\int_0^1 \frac{t^{r-1}(1-t)^{s-1}}{D(t)^m} \frac{\mathrm{e}^{vU(t)} - \sum_{j=0}^{m} \frac{(vU(t))^j}{j!}}{U(t)}\, dt = c_{r,s} \left\{ \int_0^v G_{r,s}(u)\, du - \sum_{q=0}^{m-1} \frac{G_{r,s}^{(q)}(0) v^{q+1}}{(q+1)!} \right\}. \tag{7.5}$$

The left side is an ordinary absolutely convergent integral with a removable expression at $t = t_0$. Its right side is a finite combination of classical polylogarithms and polynomial terms by (7.2). At $v = 1/2$, only $\log 2$, odd zeta values, even zeta values from the Taylor polynomial, and rational powers of π are needed.

Proof. Integrate the left side of (6.14) over $0 \le u \le v$. Uniform domination on this compact parameter segment justifies Fubini. The integral of the exponential minus its first m Taylor terms is exactly the displayed quotient. Its numerator vanishes to order $m + 1$ in U, so division by UD^m leaves a bounded function at t_0. The endpoint convergence follows from $v < s$, automatic here because $s \ge 1$ and $v < 1$. Integrating the right side proves the formula. □

Here are two explicit evaluations. In both, $U = \log(t/(1-t))$:

$$\int_0^1 \frac{\mathrm{e}^{U/2} - 1 - U/2 - U^2/8}{U(1-2t)^2}\, dt = \frac{1 - \log 2}{2}, \tag{7.6}$$

$$\int_0^1 \frac{t(1-t)}{(1-2t)^4} \frac{\mathrm{e}^{U/2} - 1 - U/2 - U^2/8 - U^3/48 - U^4/384}{U}\, dt = \frac{\log 2}{16} + \frac{3\zeta(3)}{32\pi^2} - \frac{11}{144} + \frac{\pi^2}{432}. \tag{7.7}$$

For the second identity, $G_{2,2}(u) = 1 - (1 + \pi^2/3)u^2 + O(u^4)$, so the integrated Taylor polynomial is $11/24 - \pi^2/72$ at $v = 1/2$. Combine this with (7.4) and $c_{2,2} = 1/6$. This supplies an entirely symbolic derivation of the constant.

8 A logistic Mellin transform and finite Hurwitz reductions

The centered logarithm is also a natural global coordinate. Solving (6.1) for t gives

$$t = \frac{B\mathrm{e}^U}{A + B\mathrm{e}^U}, \qquad \frac{t^{r-1}(1-t)^{s-1}}{D(t)^m}\, dt = \frac{1}{A^r B^s} \frac{\mathrm{e}^{rU}}{(1 - \mathrm{e}^U)^m}\, dU. \tag{8.1}$$

This identity holds as an ordinary change of variables away from $U = 0$. For the convergent moments $q \ge m$, it is valid across $U = 0$ as well. On $U < 0$ put $v = -U$, and on $U > 0$ put $v = U$. The two kernels become $\mathrm{e}^{-rv}/(1 - \mathrm{e}^{-v})^m$ and $(-1)^m \mathrm{e}^{-sv}/(1 - \mathrm{e}^{-v})^m$.

For an integer $m \geq 1$ and $a > 0$, define

$$\mathcal{Z}_m(z,a) = \sum_{n=0}^{\infty} \binom{n+m-1}{m-1} (n+a)^{-z}, \qquad \operatorname{Re} z > m, \tag{8.2}$$

$$\mathcal{J}_{m,a}(z) = \int_0^{\infty} v^{z-1} \frac{\mathrm{e}^{-av}}{(1-\mathrm{e}^{-v})^m}\, dv, \qquad \operatorname{Re} z > m. \tag{8.3}$$

The function $\mathcal{Z}_m$ is the equal-period Barnes-type zeta: the multiplicity counts the nonnegative integer m-tuples of sum n. We use only the explicit finite reduction proved next, not any unproved reduction of a general multiple zeta function.

Write

$$C_{m,a}(X) = \prod_{j=1}^{m-1} (X + j - a) = \sum_{k=0}^{m-1} c_{m,a,k} X^k. \tag{8.4}$$

For $m = 1$ this polynomial is one.

Proposition 8.1 (Mellin transform, order derivatives, and polygamma)**.** *For $a > 0$,*

$$\mathcal{Z}_m(z,a) = \frac{1}{(m-1)!} \sum_{k=0}^{m-1} c_{m,a,k} \zeta(z-k,a), \qquad \mathcal{J}_{m,a}(z) = \Gamma(z)\mathcal{Z}_m(z,a). \tag{8.5}$$

The first equality gives meromorphic continuation to the whole z-plane. For every $q \geq 0$ and $\operatorname{Re} z > m$,

$$\int_0^{\infty} v^{z-1} (\log v)^q \frac{\mathrm{e}^{-av}}{(1-\mathrm{e}^{-v})^m}\, dv = \partial_z^q \left(\Gamma(z)\mathcal{Z}_m(z,a)\right). \tag{8.6}$$

At an integer $p \geq m$, the non-logarithmic case is the finite polygamma expression

$$\mathcal{J}_{m,a}(p+1) = \frac{p!}{(m-1)!} \sum_{k=0}^{m-1} c_{m,a,k} \frac{(-1)^{p+1-k}}{(p-k)!} \psi^{(p-k)}(a). \tag{8.7}$$

Proof. Since $C_{m,a}(n+a) = \prod_{j=1}^{m-1}(n+j)$, expanding the polynomial in (8.2) gives the finite Hurwitz sum in its absolutely convergent half-plane. Next expand $(1-\mathrm{e}^{-v})^{-m} = \sum_{n\geq 0} \binom{n+m-1}{m-1} \mathrm{e}^{-nv}$. For real $z > m$, Tonelli and the Gamma integral prove the second identity; absolute convergence and holomorphy extend it to $\operatorname{Re} z > m$. Near zero the integrand is $O(v^{\operatorname{Re} z - m - 1})$, while at infinity it decays exponentially. Extra powers of $\log v$ are integrable on compact subsets of the half-plane, proving (8.6) by differentiation. Finally, the convergent polygamma series $\psi^{(h)}(a) = (-1)^{h+1} h! \zeta(h+1,a)$ gives (8.7). □

This proof also makes clear which parts of the construction are classical: the finite Hurwitz reduction is polynomial multiplicity expansion, and the Mellin kernel is its elementary Gamma transform. Their purpose here is to give a second derivation of the moment family and a precise Stieltjes finite-part continuation.

8.1 A second evaluation of all centered moments

For $r, s > 0$, $r + s = m \in \mathbb{N}$ and $q \geq m$ integral, (8.1) gives

$$M_q^{r,s}(A,B) = \frac{1}{A^r B^s} \left((-1)^q \mathcal{J}_{m,r}(q+1) + (-1)^m \mathcal{J}_{m,s}(q+1)\right). \tag{8.8}$$

For integer r, s this is a finite combination of generalized harmonic tails. Equating (8.8) with (6.5) proves the finite identity

$$(-1)^q \sum_{j=0}^{m-1} c_{m,r,j}\zeta(q+1-j,r) + (-1)^m \sum_{j=0}^{m-1} c_{m,s,j}\zeta(q+1-j,s)$$
$$= 2(-1)^{r+1} \sum_{k=1}^{\lfloor q/2 \rfloor} \zeta(2k)[u^{q-2k}]P_{r,s}(u). \tag{8.9}$$

In particular, replacing $\zeta(h,r) = \zeta(h) - H_{r-1}^{(h)}$ for the integer shifts shows the exact cancellation of the unwanted odd-zeta and rational harmonic tail terms. The two analytic proofs establish this at all orders; the 216 symbolic comparisons supplied with the article are independent finite consistency checks, not the reason it holds for all orders.

There is also an ordinary fractional-order extension on either half of the U-line. For example, for $\operatorname{Re} z > m$,

$$\int_0^{t_0} \frac{t^{r-1}(1-t)^{s-1}[-U(t)]^{z-1}}{D(t)^m}\,dt = A^{-r}B^{-s}\Gamma(z)\mathcal{Z}_m(z,r),$$
$$\int_{t_0}^1 \frac{t^{r-1}(1-t)^{s-1}U(t)^{z-1}}{D(t)^m}\,dt = (-1)^m A^{-r}B^{-s}\Gamma(z)\mathcal{Z}_m(z,s). \tag{8.10}$$

The powers of positive $\pm U$ use real logarithms. Their order derivatives insert $\log(-U)$ or $\log U$; no branch of a negative real power is hidden in these formulas.

9 Stieltjes coefficients, finite parts, and their primitives

The Laurent expansion convention for generalized Stieltjes constants is

$$\zeta(1+\varepsilon,a) = \frac{1}{\varepsilon} + \sum_{q=0}^{\infty} \frac{(-1)^q \gamma_q(a)}{q!}\varepsilon^q. \tag{9.1}$$

In particular $\gamma_0(a) = -\psi(a)$. All a parameters in this section are positive real numbers; differentiation may equivalently be carried out on a complex neighborhood avoiding the nonpositive integers.

For $1 \le j \le m$ set

$$\rho_{m,j}(a) = \frac{c_{m,a,j-1}}{(m-1)!}. \tag{9.2}$$

The coefficient may be zero, in which case the apparent pole is removable. Define $E_{m,j,q}(a)$ by

$$\mathcal{Z}_m(j+\varepsilon,a) = \frac{\rho_{m,j}(a)}{\varepsilon} + \sum_{q\ge 0} E_{m,j,q}(a)\varepsilon^q. \tag{9.3}$$

The finite Hurwitz reduction gives the explicit formula

$$E_{m,j,q}(a) = \frac{1}{(m-1)!q!}\left\{ c_{m,a,j-1}(-1)^q\gamma_q(a) + \sum_{\substack{0\le k\le m-1 \\ k\ne j-1}} c_{m,a,k}\zeta^{(q)}(j-k,a) \right\}. \tag{9.4}$$

The superscript on ζ here always means differentiation in its first, order variable. Formula (9.4) includes the residue-zero cases without a limiting convention.

9.1 The unnormalized Mellin finite part has an extra term

We now fix the cutoff variable to be v in (8.3), with reference split at $v = 1$. The notation

$$\mathrm{FP}\int_0^\infty v^{j-1}(\log v)^q \frac{\mathrm{e}^{-av}}{(1-\mathrm{e}^{-v})^m}\,dv$$

means the Hadamard finite part obtained by subtracting all divergent terms of the small-v expansion and discarding precisely their divergent cutoff powers and logarithms. The normalization is not inferred from the earlier t-coordinate finite part.

Theorem 9.1 (Every logarithmic finite part at a Mellin pole)**.** *For $a > 0$, $1 \le j \le m$ and $q \ge 0$,*

$$\begin{aligned}&\mathrm{FP}\int_0^\infty v^{j-1}(\log v)^q \frac{\mathrm{e}^{-av}}{(1-\mathrm{e}^{-v})^m}\,dv\\ &\quad= q!\left\{\rho_{m,j}(a)\frac{\Gamma^{(q+1)}(j)}{(q+1)!} + \sum_{\ell=0}^{q}\frac{\Gamma^{(\ell)}(j)}{\ell!}E_{m,j,q-\ell}(a)\right\}.\end{aligned} \tag{9.5}$$

In particular, the residue contribution involving $\Gamma^{(q+1)}(j)$ cannot generally be omitted.

Proof. Let

$$g(v) = \mathrm{e}^{-av}\left(\frac{v}{1-\mathrm{e}^{-v}}\right)^m = \sum_{h\ge 0} b_h v^h$$

near zero. This is analytic with $b_0 = 1$. Subtract its Taylor polynomial through $h = m - j$ on $0 < v < 1$. The remainder of $v^{j-m-1}g(v)(\log v)^q$ is integrable. Each subtracted monomial with $h < m - j$ has finite-part integral

$$\mathrm{FP}\int_0^1 v^{j-m+h-1}(\log v)^q\,dv = \frac{(-1)^q q!}{(j-m+h)^{q+1}}, \tag{9.6}$$

whereas the monomial with $h = m - j$ has finite part zero. This follows by direct integration from a cutoff δ to one; in the last case the antiderivative is $(\log v)^{q+1}/(q+1)$.

Equivalently, insert v^ε before integration. After the same subtractions the meromorphic continuation is obtained by adding $b_h/(\varepsilon + j - m + h)$. Its qth derivative in ε inserts $(\log v)^q$. The constant Laurent coefficient of that derivative is exactly the finite part calculated in (9.6). Consequently our finite part is $q![\varepsilon^q]\mathcal{J}_{m,a}(j+\varepsilon)$, where the bracket means the regular Laurent coefficient. Multiply (9.3) by the Taylor expansion of $\Gamma(j+\varepsilon)$. Its ε^q coefficient is

$$\rho_{m,j}\frac{\Gamma^{(q+1)}(j)}{(q+1)!} + \sum_{\ell=0}^{q}\frac{\Gamma^{(\ell)}(j)}{\ell!}E_{m,j,q-\ell}.$$

This proves (9.5), including any removable-pole case. □

The Gamma derivatives in this theorem are finite Bell expressions in polygamma values:

$$\Gamma^{(q)}(j) = \Gamma(j)\mathcal{Y}_q(\psi(j), \psi'(j), \dots, \psi^{(q-1)}(j)). \tag{9.7}$$

At positive integer j these reduce to γ, generalized harmonic numbers, and ordinary zeta values. Thus (9.5) is an explicit finite expression in the functions requested, at every logarithmic order.

Example 9.2 (A normalization test that detects the missing residue)**.** For $m = 2$ and $a = 1$, $\mathcal{Z}_2(z, 1) = \zeta(z-1)$ and $\rho_{2,2} = 1$. At $j = 2$, $E_{2,2,0} = \gamma$ and $\Gamma'(2) = 1 - \gamma$. Therefore

$$\mathrm{FP}\int_0^\infty \frac{v\mathrm{e}^{-v}}{(1-\mathrm{e}^{-v})^2}\,dv = 1, \tag{9.8}$$

not γ. Directly,

$$\int_\delta^\infty \frac{v\mathrm{e}^{-v}}{(1-\mathrm{e}^{-v})^2}\,dv = \frac{\delta}{\mathrm{e}^\delta - 1} - \log(1-\mathrm{e}^{-\delta}) = 1 - \log\delta + o(1).$$

For the first logarithmic moment, using $\Gamma''(2) = \zeta(2) - 2\gamma + \gamma^2$, the same theorem gives

$$\mathrm{FP}\int_0^\infty \frac{v\log v\,\mathrm{e}^{-v}}{(1-\mathrm{e}^{-v})^2}\,dv = \frac{\zeta(2)-\gamma^2}{2} - \gamma_1. \tag{9.9}$$

Here $\gamma_1 = \gamma_1(1)$ in convention (9.1).

For arbitrary $a > 0$, the same order-two calculation gives the exact polygamma finite parts

$$\mathrm{FP}\int_0^\infty \frac{v\mathrm{e}^{-av}}{(1-\mathrm{e}^{-v})^2}\,dv = 1 - \gamma - \psi(a) + (1-a)\psi'(a), \tag{9.10}$$

$$\mathrm{FP}\int_0^\infty \frac{\mathrm{e}^{-av}}{(1-\mathrm{e}^{-v})^2}\,dv = (a-1)(\psi(a)+\gamma) + \frac{1}{2} - a. \tag{9.11}$$

They follow from $C_{2,a}(X) = X + 1 - a$, $\zeta(0, a) = 1/2 - a$, and $\zeta(2, a) = \psi'(a)$. Both are in the declared v-cutoff normalization.

9.2 Parameter derivatives and exact Stieltjes antiderivatives

The multiplicity in (8.2) does not depend on a. Thus the series gives, initially in its convergent half-plane,

$$\partial_a \mathcal{Z}_m(z, a) = -z\mathcal{Z}_m(z+1, a), \qquad \partial_a^p \mathcal{Z}_m(z, a) = (-1)^p (z)_p \mathcal{Z}_m(z+p, a). \tag{9.12}$$

Meromorphic continuation proves these identities for all z. This short ladder is particularly effective at an order where a pole is multiplied by a vanishing factor.

Theorem 9.3 (A primitive for every Stieltjes coefficient)**.** *For all $m \geq 1$ and $q \geq 0$,*

$$\frac{d}{da}\left[-\frac{\partial_z^{q+1}\mathcal{Z}_m(0, a)}{(q+1)!}\right] = E_{m,1,q}(a). \tag{9.13}$$

For $2 \leq j \leq m$, define

$$F_{m,j,q}(a) = -\sum_{\ell=0}^{q} \frac{(-1)^\ell E_{m,j-1,q-\ell}(a)}{(j-1)^{\ell+1}} - \frac{(-1)^{q+1}\rho_{m,j-1}(a)}{(j-1)^{q+2}}. \tag{9.14}$$

Then $F'_{m,j,q}(a) = E_{m,j,q}(a)$. Consequently every Laurent coefficient in (9.4) has an explicit finite primitive in Hurwitz-order derivatives and Stieltjes coefficients. Definite integrals between positive endpoints are differences of the displayed primitives.

Proof. Expand (9.12) at $z = \varepsilon$:

$$\partial_a \mathcal{Z}_m(\varepsilon, a) = -\varepsilon \mathcal{Z}_m(1+\varepsilon, a) = -\rho_{m,1}(a) - \sum_{q\geq 0} E_{m,1,q}(a)\varepsilon^{q+1}.$$

Since $\mathcal{Z}_m$ is regular at $z = 0$, comparison of the ε^{q+1} coefficient proves (9.13). For $j \geq 2$, use

$$\partial_a \left(-\frac{\mathcal{Z}_m(j-1+\varepsilon, a)}{j-1+\varepsilon} \right) = \mathcal{Z}_m(j+\varepsilon, a).$$

The regular ε^q coefficient on the left is (9.14); its final term comes from the residue at $j-1$. Coefficient comparison proves the derivative assertion. All expansions are locally uniform in $a > 0$, as follows either from the finite Hurwitz representation or from analyticity before continuation. □

For $m = 1$, $\mathcal{Z}_1(z, a) = \zeta(z, a)$ and $E_{1,1,q} = (-1)^q \gamma_q(a)/q!$. Thus the familiar Stieltjes primitive appears as the first member of the ladder:

$$\int \gamma_q(a)\, da = \frac{(-1)^{q+1}}{q+1} \zeta^{(q+1)}(0, a) + \text{constant}. \tag{9.15}$$

For $m = 2$,

$$\mathcal{Z}_2(z, a) = \zeta(z-1, a) + (1-a)\zeta(z, a), \tag{9.16}$$

so (9.13) gives the nontrivial weighted example

$$\frac{d}{da}\left[-\frac{1}{2}\left\{ \zeta''(-1, a) + (1-a)\zeta''(0, a) \right\} \right] = \log\Gamma(a) - \frac{1}{2}\log(2\pi) - (1-a)\gamma_1(a). \tag{9.17}$$

We used $\zeta'(0, a) = \log\Gamma(a) - \frac{1}{2}\log(2\pi)$, with the standard positive-real normalization [8]. More generally,

$$\frac{d}{da}\left[-\frac{\zeta^{(q+1)}(-1, a) + (1-a)\zeta^{(q+1)}(0, a)}{(q+1)!} \right] = \frac{\zeta^{(q)}(0, a) + (1-a)(-1)^q \gamma_q(a)}{q!}. \tag{9.18}$$

This is an all-order identity, not an extrapolation from its first few numerical values. Higher m yields polynomially weighted Stieltjes combinations with a finite list of shifted Hurwitz-order derivatives.

10 Corrections, compatibility, and integration recommendations

10.1 Targeted audit outcome

No mathematical error was found in the common-interval mixture or the integer-order partial-fraction theorem of the inspected section [1]. The present result should be added as a continuation, not used to replace their statements with an alleged correction. Its hypotheses are different: arbitrary nonintegral real twists, but positive integer total order for the finite-part formula. The source's unrestricted fractional *total* order in a common interval is retained.

Three tempting extensions are rigorously rejected in this article. First, removing the common-interval condition while using an uncorrected principal value is contradicted by (4.5). Second, inferring vanishing order derivatives from a vanishing integer-tuple correction is contradicted by (5.8). Third, replacing the finite part of $\Gamma(z)\mathcal{Z}_m(z, a)$ by Gamma times the regular part of $\mathcal{Z}_m$ omits a residue term; the independent cutoff calculation in (9.8) detects the error immediately. These are statements about candidate extensions and normalization rules. They are not attributed to the current manuscript without textual evidence.

10.2 Compatibility with the existing proof chain

At a common integer chamber, Theorem 4.1 reduces to the manuscript's beta theorem specialized to integer total order. At a positive integer tuple, it becomes an affine finite-part interpretation of the same convolution already given by the source's finite resolvent identities. These are consistency checks at the overlap, not separate period conjectures. The harmonic Bell weights in (5.6)–(5.7) are compatible with the finite-product and Gamma-coefficient conventions in the harmonic section [3]; we do not claim to re-prove a previously unresolved Choi case.

The name "correction" is used in two different senses that must stay separate in integration. The sine polynomial corrects a beta mixture across integer *twists*. The Gamma-derivative term corrects a *Laurent coefficient extraction* through a pole. Neither is an unannounced Dirac distribution in the spatial variable. Furthermore, the ordinary moment identities and the subtracted generators must not be relabeled as regularized integrals: their vanishing numerators make them genuinely convergent.

10.3 Suggested placement and review status

The mathematical core is supplied as `integration/07-beta-crossing-identities.tex`, with namespace `bc:` for all labels and a separate small preamble fragment. Its natural placement is after `07-mixed-twist-identities.tex` in the Hurwitz–Stieltjes material. The Mellin and primitive subsections can instead be cross-referenced from the integration/differentiation chapters. A machine-readable status ledger lists the new statements, their proof locations, overlap with classical ingredients, and verification coverage.

The supplied proofs are mathematical arguments, not proof-assistant formalizations. They have not undergone independent peer review. The replay scripts test finite algebra and selected numerical instances; they do not certify every continuum statement. The generated package changes no repository files. It is intended for normal analytic audit and integration, preserving the source's existing status distinctions.

11 Further research questions and proposed next steps

The following questions are deliberately phrased as open tasks rather than conclusions inferred from the finite tests.

1. **Noninteger total order across chambers.** For $\sum \alpha_j \notin \mathbb{N}$, determine a comparably useful explicit real-boundary or finite-part mixture. The right-half-plane regularized Dirichlet formula remains available, but the two boundary distributions are no longer a real rational finite part plus a single delta derivative. A successful extension should specify the global branch convention and recover (4.2) at every integer total order.

2. **Unbalanced order derivatives.** Allow $\sum_j v_j \neq 0$, so that differentiation changes the total order. This introduces order derivatives of the intermediate kernel and logarithmic boundary distributions at the crossing. Determine the exact replacement for the finite polynomial jet formula, with all local counterterms specified. The balanced theorem does not settle this case.

3. **An explicit connection to the manuscript's Stieltjes jets.** Translate the balanced derivatives into the exact normalization of the manuscript's twisted Stieltjes families. Identify

which Bell combinations reduce to finite single-Hurwitz expressions and which require the existing convergent unequal-shift expansions. A finite spectral correction alone does not prove the desired finite reduction of the full convolution.

4. **Moving twists and chamber transitions.** Differentiate in the twists as a vertex approaches an integer, keeping track of the divergent individual Fourier coefficient. Determine canonical subtractions and transition identities that agree from both neighboring chambers. At a fixed nonintegral twist configuration our formulas are explicit, but uniform passage through an integral vertex needs a separate theorem.

5. **Rational endpoint polylogarithm reductions.** Formula (7.2) evaluates the centered primitive at $v = p/q$ through polylogarithms at qth roots of unity. Develop a conductor-aware reduction of the full family, including its Taylor subtractions, into the canonical cyclotomic basis used elsewhere in the repository. At $q = 2$ this article provides the finite odd-zeta formula. No claim of minimal depth or period independence is implied.

6. **Repeated parameter primitives.** Iterate the construction behind (7.5). At the hth primitive the natural integrand contains an exponential remainder divided by U^h and a correspondingly larger subtraction polynomial. Give a single closed coefficient formula, branch normalization, and exact root-of-unity endpoint evaluations at every h.

7. **Mellin residues with unequal periods.** Replace $(1-\mathrm{e}^{-v})^{-m}$ by a product $\prod_j (1-\mathrm{e}^{-\omega_j v})^{-1}$. For commensurable positive periods, find the precise finite Hurwitz decomposition, including quasipolynomial multiplicities and every logarithmic finite part. For incommensurable periods, determine which pieces of the Stieltjes primitive ladder survive without pretending that the same finite one-variable reduction holds.

8. **Coordinate-change laws for finite parts.** Derive an explicit all-order local formula comparing the affine t-finite part with the logarithmic U-finite part for nonconvergent moments. The ordinary $q \geq m$ identities already agree automatically. The remaining finite discrepancy should be expressed in the finite jet of the logistic map, rather than absorbed into an unspecified constant.

9. **Formal proof and exact regression coverage.** Formalize the scalar boundary-value theorem and chamber-rank statement, then use them as interfaces for the Fourier and special-function calculations. In the short term, add exact multiparameter Bell and higher-pole primitive tests, together with independent interval quadrature for selected convergent integrals. Neither failed numerical searches nor a passing finite suite should alter open period statuses.

A A direct finite proof of the two-point density derivative

The local derivative in the two-vertex case also has a direct algebraic proof, independent of boundary-value evaluation. Let $\alpha + \beta = m$ and $N = m - 1$. For $0 < t < 1$,

$$\frac{d^{m-1}}{dt^{m-1}}\left(t^{\alpha-1}(1-t)^{\beta-1}\right) = \left(\prod_{h=1}^{m-1}(\alpha-h)\right) t^{-\beta}(1-t)^{-\alpha}. \tag{A.1}$$

Indeed Leibniz's rule, after extracting $t^{-\beta}(1-t)^{-\alpha}$, gives

$$\sum_{j=0}^{N}\binom{N}{j}(\alpha-1)^{\underline{j}}(\beta-1)^{\underline{N-j}}(-1)^{N-j}t^{N-j}(1-t)^{j}.$$

Here the underlined exponent denotes a falling factorial. Since $\beta = m - \alpha$, each product of falling factorials and its sign is $\prod_{h=1}^{N}(\alpha - h)$, independent of j. The remaining sum is $(t + (1-t))^N = 1$, proving (A.1).

For the affine pushforward $y = B - (A+B)t$, its density is

$$h(y) = \frac{\Gamma(m)}{\Gamma(\alpha)\Gamma(\beta)(A+B)} \left(\frac{B-y}{A+B}\right)^{\alpha-1} \left(\frac{A+y}{A+B}\right)^{\beta-1}.$$

Use (A.1) and the chain rule at $y = 0$. The powers of $A + B$ cancel. Gamma reflection then yields

$$h^{(m-1)}(0) = \frac{\Gamma(m)}{\pi A^{\alpha} B^{\beta}} \sin(\pi\alpha),$$

which is exactly (3.5). Apparent singularities in intermediate Gamma quotients at integer parameters are removable; the polynomial identity itself has no such restriction. The exact verification script checks its finite Leibniz expansion through $m = 12$.

B Reproducible verification and evidence boundaries

The scripts require Python, SymPy and mpmath. The recorded run used Python 3.13.5, SymPy 1.14.0 and mpmath 1.3.0. All numerical diagnostics were run at 65 decimal digits. The following commands can be run from the package directory:

```
python verification/check_exact.py
python verification/check_numeric.py --dps 65
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The two scripts write JSON receipts next to themselves and stop on a failed assertion. Their numerical tolerances are explicitly recorded per test. These tolerances are diagnostics, not rigorous error enclosures.

B.1 Finite exact checks

Family	Checks
Two-point finite density derivative	12
Multiplicity polynomial	12
Polynomial derivative recurrence	12
Individual coefficient derivative recurrence	66
Sine/exponential jet versus two powers	14
Cosine/exponential jet versus two powers	14
Chamber rank over exact rational twists	120
Mellin moments versus cotangent coefficients	216
Gamma prefactor through a simple pole	11
Polylogarithm primitive telescoping	25
Explicit phase and normalization constants	3
Rejected uncorrected formulas	3
Total	508

The 216 moment comparisons cover $1 \le r, s \le 6$ and the six orders $m \le q \le m+5$ for each pair. They treat the zeta values as independent formal symbols and use exact rational harmonic sums, so accidental numerical cancellation cannot hide a coefficient error. The 120 rank checks use rational twists and exact matrix ranks, with a fixed seed. The proof of Theorem 5.3, not this finite sample, supplies the general rank statement.

B.2 Floating-point diagnostics

Comparison	Cases
Direct affine beta finite parts	6
Original-variable ordinary centered moments	27
Polylogarithm primitive versus quadrature	16
Half-period primitive evaluation	1
Ordinary inverse-logarithm integrals	2
Mellin integrals and first order derivatives	24
Direct subtracted logarithmic Mellin finite parts	12
Stieltjes primitive differentiated in the parameter	6
Total	94

The beta finite parts are calculated by Taylor subtraction in the original affine coordinate, not by fitting values at a small imaginary regulator. The moment diagnostics integrate in t, independently of the Mellin substitution. The pole diagnostics subtract the small-v series, integrate the remainder, and add the exact monomial finite parts. These are meaningfully different evaluations from the closed forms, although floating-point quadrature and finite Taylor arithmetic still do not produce rigorous intervals.

The supplied receipts report all 508 exact checks and all 94 diagnostics as passing. Near a subtraction point the numerical script uses a finite Taylor series to avoid catastrophic cancellation. That implementation detail reinforces why the diagnostics are not a replacement for the analytic Taylor-remainder arguments in the article. A displayed zero floating-point residual means agreement at the working precision, not an independently certified exact equality.

B.3 Source and artifact provenance

The source manifest records the observed reference commit, the targeted chapter blob, the scope of the other reads, and the names and Git blob identifiers of the six incoming archives that were listed but not opened. It does not assert that all moving-branch reads were a single atomic snapshot. The targeted mathematical source was re-read at the reference commit, eliminating that ambiguity for the theorem being extended.

The package includes the complete LaTeX source, its compiled PDF, an integration fragment extracted from the mathematical sections, the two verification scripts, their JSON receipts, a status ledger, build instructions, and SHA-256 checksums. Build by-products are excluded from the distribution. The PDF was rendered and visually inspected as part of package preparation; the separate artifact inspection record documents that review. No proof-assistant or external peer-review status is inferred from the artifact checks.

References

[1] ProveIt Contributors, *Unequal twists through beta mixtures and finite resolvent identities*, `07-mixed-twist-identities.tex`, in the manuscript chapters directory, reference commit `446dca9bd65c20e409d1d2561d4ffad408103e2a`. Pinned GitHub source.

[2] ProveIt Contributors, *Polylogarithms and their Arithmetic Bridges*, manuscript README, inspected during this continuation. The moving-branch reading is recorded in the provenance manifest. GitHub manuscript directory and README.

[3] ProveIt Contributors, *Gamma coefficients prove every harmonic-order identity*, `07-harmonic-gamma-coefficients.tex`, inspected source blob `1b18b485517cd551f6a938b1b9736cea942eacf5`. GitHub harmonic-coefficient section.

[4] NIST Digital Library of Mathematical Functions, §5.12, *Beta Function*. `https://dlmf.nist.gov/5.12`.

[5] NIST Digital Library of Mathematical Functions, §5.5, *Functional Relations* for the Gamma function. `https://dlmf.nist.gov/5.5`.

[6] NIST Digital Library of Mathematical Functions, §5.15, *Polygamma Functions*. `https://dlmf.nist.gov/5.15`.

[7] NIST Digital Library of Mathematical Functions, §25.12, *Polylogarithms*. `https://dlmf.nist.gov/25.12`.

[8] NIST Digital Library of Mathematical Functions, §25.11, *Hurwitz Zeta Function*. `https://dlmf.nist.gov/25.11`.

[9] H. A. Weldon, *Mishaps with Feynman parametrization at finite temperature*, Physical Review D **47** (1993), 594–600. doi:10.1103/PhysRevD.47.594. Abstract consulted for context; no theorem in this article depends on an uninspected derivation in that paper.

[10] P. F. Bedaque and A. Das, *Feynman parametrization and the degenerate electron gas*, Physical Review D **47** (1993), 601–607; arXiv:hep-ph/9209273. doi:10.1103/PhysRevD.47.601. Abstract consulted for context.